



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA
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91258 / B0385 Natural Language Processing

Lesson 3. Vector Space Model

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Current Status

Current Status

You know...

- what is natural language processing
- there are two main paradigms: rule-based and statistical

On your own, you have...

- set up a Python development environment
 1. command line
 2. PyCharm or any other option (e.g., Eclipse)
 3. Google's Colab

- played with `spacy` (and `nlTK`)

On your own, you (could) have...

- played with `pandas`
- found out what is **git** (and perhaps $\text{\LaTeX}$ as well)

You can...

- open a text file (Python intro)
- tokenise and normalise text
- build some text representations

Representations Revisited

Representations Revisited

1. Use NLTK¹, Spacy², or anything else to tokenise
2. Use `.lower()` to casefold (ignore capitalisation)
3. Use Porter's stemmer to drop suffixes
or use a lemmatiser to find the *actual* root of words
4. Discard stopwords from the text*
5. Build a vectorial representation*

¹<https://www.nltk.org>

²<https://spacy.io>

Stopwords

Common words in a language that occur with a high frequency, but carry much less substantive information about the meaning of a phrase (Lane et al., 2019, p. 51–54)

Alternative 1 Consider the most frequent tokens in a reference corpus as stopwords (remember Genesis from P4P?)

Alternative 2 Take an existing list of stopwords³

en	es	it
i	a	altri
me	ahora	certa
my	allí	della
it	cerca	nessuna
is	el	prima
do	es	quello
the	unas	solito
will	vez	va
other	yo	via

³For instance, from NLTK, sklearn, or <https://github.com/stopwords-iso>

Stopwords

Discarding stopwords

- They are the most frequent tokens in the documents
- Discarding them significantly reduces the computational effort
- Typical size of a stopwords list: a few hundred words
- For some applications (e.g., **topic clustering**), they can be safely discarded
- For some others (e.g., **dialogue**) they cannot

Stopwords have to be considered with a grain of salt
(as everything in NLP)

Vector representation

BoW

- A text is represented as the bag (set) of its words
- It disregards grammar
- It disregards word order
- It (can) consider frequency

From (Lane et al., 2019, p. 41)

More Basic Algebra

x and y



<https://twitter.com/miniapeur/status/1710074831079690394>

Dot product

Algebraically, it is the sum of the products of the corresponding entries of the two sequences of numbers $a \cdot b$

$$\begin{aligned} a \cdot b &= \sum_{i=1}^n a_i b_i \\ &= a_1 b_1 + a_2 b_2 + a_3 b_3 + \cdots + a_n b_n \end{aligned}$$

```
a = [1,2,3]
b = [3,4,6]
my_sum = 0
for i in range(len(a)):
    my_sum += a[i] * b[i]
```

There are better —more efficient— ways to compute a dot product
Now, we can use the dot product to compare two documents ($\sim$ similarity)

Vector space model

“[...] an **algebraic** model for representing text documents (or more generally, items) as vectors [...]”⁴

Some applications

- Relevance ranking in keyword-based search
- Document clustering to “discover” structure and relations in a text collection

(not SotA for most tasks, but it represents a **minimum viable product**)

</> Let us see it working

⁴https://en.wikipedia.org/wiki/Vector_space_model

Next time...

VADER

References

Lane, H., C. Howard, and H. Hapkem
2019. *Natural Language Processing in Action*. Shelter Island, NY:
Manning Publication Co.